

EEG based interpretation of human brain activity during yoga and meditation using machine learning: A systematic review

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ABSTRACT

Objectives: The present investigation is to study the impact of yoga and meditation on Brain waves concerning physical and mental health. There are mainly three stages (steps) in the brain wave classification: (i) pre-processing, ii) feature extraction, and iii) classification. This work provides a review of interpretation methods of Brain signals (Electroencephalogram (EEG)) EEG during yoga and meditation. Past research has revealed significant mental and physical advantages with yoga and meditation.

Methods: The research topic reviewed focused on the machine learning strategies applied for the interpretation of brain waves. In addressing the research questions highlighted earlier in the general introduction, we conducted a systematic search of articles from targeted scientific and journal online databases that included PubMed, Web of Science, IEEE Xplore Digital Library (IEEE), and Arxiv databases based on their relevance to the research questions and domain topic. The survey topic is relatively nascent, and therefore, the scope of the search period was limited to the 20-year timeline that was deemed representative of the research topic under investigation. The literature search was based on the keywords “EEG”, “yoga” and “meditation”. The key phrases were concatenated using Boolean expressions and applied to search through the selected online databases yielding a total of 120 articles. The online databases were selected based on the relevancy of content with the research title, research questions, and the domain application. The literature review search, process, and classification were carefully conducted guided by two defined measures; 1.) Inclusion criteria; and 2.) Exclusion criteria. These measures define the criteria for searching and extracting relevant articles relating to the research title and domain of interest.

Results: Our literature search and review indicate a broad spectrum of neural mechanics under a variety of meditation styles have been investigated. A detailed analysis of various mental states using Zen, CHAN, mindfulness, TM, Rajayoga, Kundalini, Yoga, and other meditation styles have been described by means of EEG bands. Classification of mental states using KNN, SVM, Random forest, Fuzzy logic, neural networks, Convolutional Neural Networks has been described. Superior research is still required to classify the EEG signatures corresponding to different mental states.

Conclusions: Yoga practice may be an effective adjunctive treatment for a clinical and aging population. Advanced research can examine the effects of specific branches of yoga on a designated clinical grouping. Yoga and meditation increased overall healthy brain activity.

1. Introduction

Yoga and meditation have been used from ancient times as standard practices of alternative medicine [3] to bring a stress-free life [4]. The popularity of yoga has risen in recent years as not just recreational exercise but as a means of reducing stress and anxiety [2], increasing

physical fitness, and improving mood and overall well-being [5–7]. Yoga has been demonstrated to have several positive effects on the cardio-respiratory condition [1]. Multiple studies provide evidence that yoga can increase cardio-respiratory efficiency and metabolic process [8,9]. Six months of yoga practice can bring the heart rate and Blood Pressure (BP) to the normal level [8]. Yoga has been also beneficial for

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individuals with metabolic conditions such as Type-2 diabetes and obesity, as it increases glycemic control and nerve conduction velocity, decreases Body Mass Index (BMI) and total serum cholesterol level [10, 11]. In addition to the cardio-respiratory and metabolic improvements, yoga activity correlated with musculoskeletal benefits. Weight-bearing yoga activity can attenuate bone re-absorption and decreases the risk of bone density in menopausal women [12]. The Yoga activity is mostly used as a form of exercise, as it requires the involvement of various muscles. One meditation practice, named Mindfulness, has already been applied in a variety of clinical problems [13–15].

Yoga asanas involve different body postures that the individual executes. Each asana may also be held isometrically for an allotted amount of time or breath cycles. Vinyasa, Bikram, kundalini, and hatha yoga are different types of yoga in the category of asana based yoga. These models can be practiced individually or in classroom learning. Many people consider yoga a sort of meditation. The meditation-based practice involves a purposeful relaxation of the mind through the separation of thoughts and/or concentration on one's own breathing. Meditation commonly worked on taking a seat aside for breathing, does not require any dynamic movements. The third sort of yoga practice is a breathing-based practice. This training involves purposeful inhalations and exhalations at a designated speed and intensity, referred to as pranayama. Pranayama has different breathing practices, which incorporate Sudarshan kriya and Bhastrika pranayama. Breathing-based exercises are also practiced in a quiet and seated position and can be practiced on an individual or classroom basis [16,17].

The above three categories of yoga can be consolidated into one session or rehearsed independently. Asana, meditation, and breathing-based practices elicit various and specific effects on cognitive and neurological functions. With the positive impact of yoga on the body, recent research has begun to explore the cognitive effects of yoga. Previous qualitative studies have shown positive effects of yoga on depression and anxiety [18,19]. Several studies have also shown an increase in cognition after yoga in practitioners versus non-practitioners. Research evaluating 108 school children between 10 and 17 years of age tested spatial and verbal memory before and after a 10-day regimen of pranayama learning. The results showed an increase in spatial and verbal memory power after the Sudarshan Kriya yoga [20]. A study examining adjunctive treatments for bipolar patients have shown a high rate of self-reported cognitive benefits as a result of yoga practice.

Many subjects with bipolar depression reported that ongoing yoga practice assisted their focus and sense of acceptance [21]. Often linked to cognitive benefits, another study showed that six weeks of hatha yoga enhanced the capacity of memory and attention-switching ability in older adults [22]. Breathing-based yoga, in the form of fast and slow pranayama practice, has also shown improvement in cognitive performance in the domains of reaction time [23]. After 12 weeks of pranayama practice, 84-healthy adult participants significantly improved psychometric test scores, including letter cancellation tests, A and B trail tests, forward and reverse digit periods, and red light and green light auditory and visual reaction times. The improvement in cognitive performance from yoga practice and interventions is evident, but the mechanisms behind this improvement remain unclear. Changes in cognition are often a result of changes in neuronal activity, structural activation [24] within the brain. Understanding what can elicit changes that are occurring within the brain leading to improved cognition, can give insight into the development of cognitive interventions in both healthy and clinical populations.

Previous research shows that the EEG signal changes can be observed during meditation and are different from normal people [25]. The aim of this review is, therefore, to investigate the particular neural signal changes during yoga and meditation practice that may improve mental health and overall well-being. EEG signal classification (normal and meditation people) involves three main steps: pre-processing, feature extraction, and data classification. The contribution of this paper is: (a)

to systematically review the pre-processing techniques after acquiring EEG. (b) To review the feature extraction techniques using time and frequency domain techniques (c) To review the functional connectivity during meditation state for theta, alpha, and beta frequency bands and to compare the performance of different classifiers. The survey includes signal processing (Sections 3, 4 and 5) in which Pre-processing, extraction of features, and classification briefly outlined. Finally, we addressed the current state of the art in the identification of brain states based on the EEG signal shown in Table 2.

2. Background

Literature already exists on the meditation-induced changes in EEG brain waves. A study revealed that meditation induces electrical changes in the brain vary among five different traditions of meditation like Tibetan Buddhists (TB), Qigong Yoga, Sahaja Yoga (SY), Ananda Marga Yoga (AY), and Zen [26]. The results of these five types of meditation have many impacts, particularly on delta and beta bands, as significant changes observed in these bands. Upon meditation, in the TB group, delta band connectivity was noted posteriorly from left to right, while in the AY group, connectivity was noted from anterior left to posterior right. These conclusions are not always consistent. For example, in Qigong, Chinese meditation technique includes a large number of connections. However, this research shows the standard features in five traditions of meditation but there was no control group. Analysis of principal functional connectivities during the delta band reflected the interdependence between different functions that are lowered with practicing meditation irrespective of meditation style. The inhibitory and excitatory (delta and beta band activities) brain connectivities show this reduction [27].

In another study, meditation-induced changes of two meditation styles, mindfulness meditation, and loving-kindness (or Metta) meditation, was investigated [28]. The authors discussed the effects of pre-frontal alpha asymmetry in EEG activity. Subjects low in brooding showed a response to loving-kindness meditation while the opposite was observed for subjects with high brooding. Comparisons to rest group showed useful state effects of both the styles of meditation. It accounts for their clinical use for previously depressed patients, although various limitations like unspecific factors, small sample size, and novice subjects were observed in the study. The study previously carried on depressed patients. Meditation has not only shown benefits to symptoms sometimes it can have adverse effects, such as epileptogenesis and mood swings [29]. Concerning the use of relaxation for high BP, there is a need for more randomized clinical trials that could provide some data [30]. In people with neurophysiological disabilities, the positive effects of EEG driven meditation may turn out to be negative. Deep meditation causes gamma-band bursts of high-frequency [31]. In resting interictal EEG (the period between Epilepsy seizures), patients with generalized epilepsy display increased gamma band activity, especially 30–50 Hz [32], [33]. Changes in the epilepsy signal have been theorized in Trans Meditation (TM), but some people dismissed these arguments. Meditation predisposing (increasing risk of) Epileptic seizures is a controversial topic needing more scientific study. Paroxysmal gamma (PG) was observed during the Bhramari Pranayama (BhP) [34]. High frequencies in beta (12–30 Hz) and gamma (> 30 Hz) and enhanced theta intensity with complex Morlet wavelets were observed with Fourier analysis. Since, epilepsy also exhibits similar spiky shape and EEG patterns in the temporal lobe, as seen in BhPr, it may reflect the epileptic activity. Another study has brought light to TM as aggravating or treating epilepsy [35]. The steps for EEG-based studies usually record the subject's brain activity or physiological signals, extract time domain, frequency domain, or time-frequency domain features of these signals and classify the useful features through different classifiers. This study planned to explore signal processing and machine learning approaches in grading brain wave patterns during Yoga and meditation, as shown in Fig. 3.

3. Methods

The literature that was chosen for this review began by searching for the terms “yoga” and/or “meditation” with the terms “EEG”, “brain”, “brain waves”. The databases used, to search for studies with these terms, included PubMed, Web of Science, IEEE explore, Google Scholar, and EBSCO host. Potential articles were categorized into groups including, quantitative research, qualitative research, and single case studies. The inclusion criteria consisted of quantitative studies that were conducted between the years 1990–2019 that were published in journals, and examined either the effects of a yoga intervention or differences between yoga practitioners and non-practitioners. The inclusion criteria was narrowed to studies that examined posture (asana), breathing, and meditation-based yoga and used brain waves, brain changes in brain structure as outcome variables. The exclusion criteria was composed of studies conducted before the year 1990, studies not written in English, unpublished work, or articles based on an single individual’s opinion. After screening for inclusion and exclusion criteria, 50 studies were reviewed. These studies were then further categorized based on the outcome variables measured, into the groups: “Brain Waves”, “pre-processing”, “feature extraction” and “classification”.

3.1. Pre-processing technology

3.1.1. Data collection and noise removal

EEG electrodes for acquiring data are placed into six scalp regions namely (frontal, central, parietal, occipital and temporal) as shown in Fig. 1.

Previous researchers used different EEG data collecting devices for their research as shown in Fig. 2. The studies examined posture (asana), breathing, and meditation-based changes and brain structural activation, as outcome variables. Asieh Ahani et al. [36] acquired EEG data of 34 meditators using BioSemi active device. Lee et al. [37] used Nicolet clinical EEG acquiring system with a 250-Hz sampling rate. Thilo Hinterberger et al. [38] utilized cap, where electrodes are placed using 64 channel ANT Waveguard, and placed according to the international 10-10 system. Sharma et al. [39] used a 64 channel waveguard cap with

the international 10/20 system. Narendra Jadhav et al. [25] collected signals with 14 electrode EMOTIV EEG hardware. EEG acquired during four emotions Happy, Angry, Sad, and Relax pre and post-meditation state was obtained. Jim Lagopoulos et al. [40] recorded EEG during relaxation and meditation using 20 electrode system with the international 10/20 method with the help of a Neuroscan cap. Chang et al. [41] collected EEG from 8-channel SynAmps amplifiers of 45 min duration. Manish Sagar et al. [42] used an FMS electrode cap of 88 electrodes to collect EEG data during focused attention. Atefeh Goshvarpour [43] et al. utilized 16 channel Power lab to collect EEG data concerning the 10/20 electrode system. Mohammed ShafiquIslam [44] et al. removed motion artifact using independent component analysis. This method shows an average increase of 9% in the accuracy of detection.

Eglė Butkevičiūtė et al. [45] used sparsity and BEADS filter to remove noise in the biomedical signals like ECG and EEG. Rekh Ram Janghel [46] et al. utilized different filters like Recursive Least Squares (RLS), Shift Moving Average (SMA) and Least Mean Squares (LMS) to remove noise in the EEG signals and these filters improved accuracy in the detection of Epilepsy. Petr Nejedlý [47] et al. removed artifacts from intra-cerebral EEG signals using convolution neural networks. Xiao Jiang [48] et al. reviewed various filtering techniques to remove the artifacts in the EEG signal. Ghazaleh Darmani [49] et al. used Independent component Analysis (ICA) and high pass filtering to remove noise and studied the effect of anti-epileptic drugs on the human brain.

3.2. Different brain wave patterns on EEG and meditation styles

3.2.1. Brain waves

An EEG electrode mostly senses the electrical activity in the brain region. Nonetheless, thousands of neuron contribute to the electrical signal captured by the electrodes. There are more than 100,000 neurons in one square millimeter of the cortex. It is only when the activity of a region is associated with simultaneous electrical activity that one begins to obtain differentiated wave shapes and frequencies in the form of EEG. Alpha, beta, delta, and theta are four primary periodic rhythms reported in EEG [50,51]. The frequency (Hz or cycles/sec) and amplitude define these rhythms. The amplitudes of scalp electrodes are in the micro volt range.

A detailed description of assigned EEG band [52,53] has been given in Table 1 and Fig. 4.

Brain wave activation represents the electrical activity of neurons, specifically the voltage fluctuations from ionic movement within neurons in the brain. This electrical activity is recorded using the Electroencephalogram (EEG), and the EEG will characterize this electrical activity as waves or oscillations. These oscillations are representative of specific neuronal events throughout the brain. Brain activity reflected in these waves is naturally taking place in both active and resting states. However externally stimulation can additionally elicit these waves. Repetitive transcranial magnetic stimulation, transcranial direct current stimulation, and transcranial alternating current stimulation are a number of standard methods of obtaining and altering brain waves. These techniques are used in both clinical and research settings to help verify the integrity of and higher understanding of the central nervous system. These three strategies can assist in modulating and influence current brain rhythmic activity and elicit specific brain wave types.

EEG recordings begin with the placement of an array of electrodes across the human scalp. EEG will then measure the voltage fluctuations from ionic flows of neurons in the brain. EEG detects these fluctuations and represents them as a wave, or oscillation, and notes at which times they are activated and at which frequency.

3.2.2. Delta waves

These waves are the slowest brain waves having a frequency range between 1 and 4 Hz. Delta waves are observed in deep sleep and having a high amplitude and the slowest waves (low frequency). These waves can only be generated when the person is in deep meditation state or in

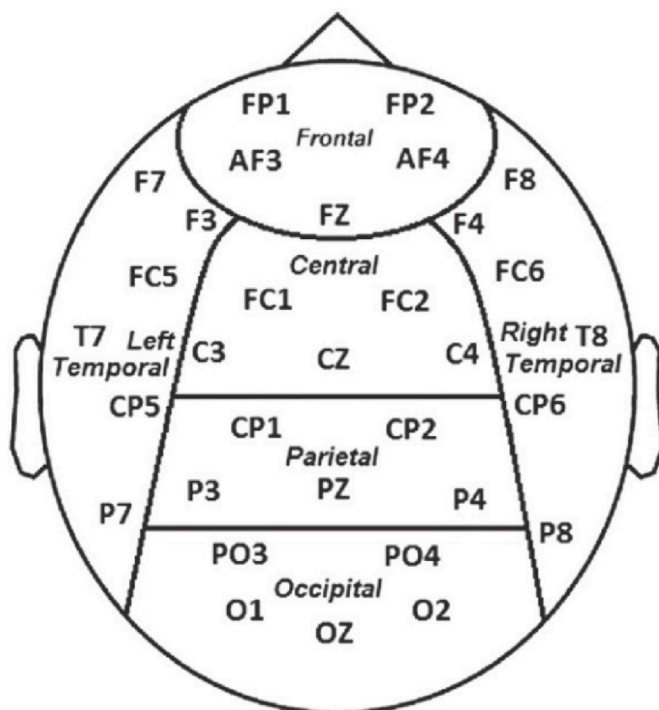


Fig. 1. Regions of scalp.

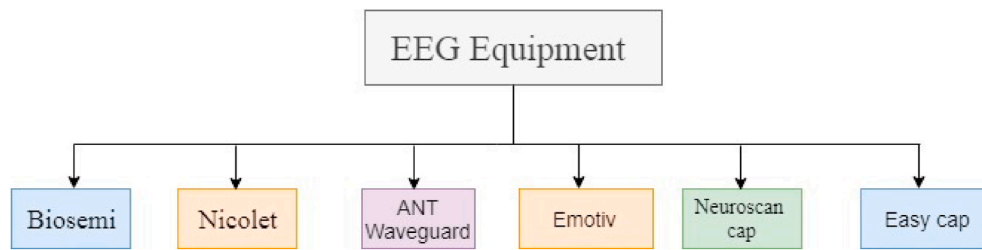


Fig. 2. Different EEG acquisition systems.

Table 1

EEG frequency bands.

Frequency-bands	Brain-regions	Characteristic feature
Delta (<4hz)	i)Children- Posterior (ii)Adults-Frontal	i)Inclination towards the greatest voltage (ii)In babies up to one year appeared during profound sleep stages in adults (iii)Slow brain patterns (iv)Minimum brain self-conscious involvement
Theta (4–8hz)	Medial prefrontal and anterior cingulate cortex	(i)Voltage from 30 to 60 μ V (FmTheta) (ii)In ordinary kids and babies, relaxed condition in adults during somnolence, sleep, and meditation. (iii)Neural indicator of inner processes requiring self-management.
Alpha (8–13hz)	i)High voltages on leading side, posterior on each side of the brain with high voltages in the lead side (ii)Slower in the anterior position and faster in the posterior position,	i)Voltages μ 50 μ V (ii)Predominant frequency in adults (iii)Tasks requiring attention, semantic memory (iv)Easy to see with closed eyes and mentally inactive. (v) ERP:Alpha is desynchronized and theta is synchronized during broader task requirements. (vi) Beta rhythms of fast low amplitude dominate the waves. Alpha blocking happens when eyes are opened in a well-lighted space.
Beta (13–30hz)	(i) Cerebral Posterior occipital in infants and frontal areas for grown up (ii) Frontocentral	(vii) μ rhythm: Alpha inaction in sensori-motor cortex (i) Practically amplitude μ 30 V. (ii) β is a quick process that occurs when persons are alert, attentive, engaged in problem solving, assessing, making decisions, or concentrated mental activity.
Gamma (>30hz)	Different brain regions	(iii) Beta bursts move the system into attention state allowing gamma coordination. (i) Increases in focused stimuli. (ii) Provides long-term memory synaptic plasticity by closely monitoring regional blood flow changes. (iii) Higher in focused stimuli.

dreamless sleep shown in Fig. 3. These waves show an inclination towards the greatest voltage. In babies up to one year appeared during profound sleep stages. In adults it shows slow brain patterns and minimum brain self-conscious involvement.

3.2.3. Theta waves

Theta waves (4–8 Hz) [54], also known as theta rhythm, occur once a repetitive task becomes autonomous, hardly requiring any focus to be completed. In essence the theta state is established once the task(s) are completed in a recognizable routine. A study examining this, presents a review regarding studies which utilized EEG stimulation to indicate theta activity with repetitive tasks [54]. Theta waves can occur in both cortical and hippocampal regions. When stimulated, theta waves, similar to alpha waves, have been shown to reduce anxiety. Theta waves may play a role in the function of short-term memory, according to studies which [55] reported the most prominent theta activity in the hippocampus, suggesting theta waves may influence the process of building memories.

3.2.4. Alpha waves

Alpha waves are neural oscillations at the frequency of 8–12 Hz [56] that are found within the cortex, occipital lobe, and thalamic regions. Typically, alpha waves are detected via EEG. Alpha waves typically have large amplitudes and occur during moderate levels of brain activity [57]. Specifically, Alpha waves occur while and individual is temporarily idle, but still alert. It is typical for significant amounts of these waves to occur during a sleeping or drowsy state. Functionally, alpha

waves inhibit areas of the cortex and play a vital role in networking between neurons. Alpha brainwave activity has been correlated with physiologic behavior such as decreased degree of pain and discomfort. The Alpha frequency was also shown to be correlated to cognitive performance, including the speed at which information is retrieved from memory. A positive correlation between fast and accurate memory performance and alpha frequency was found. Alpha wave activity may also improve word recognition in older adults as well as facilitate working memory [56]. Along with cognitive benefits, elevations in alpha wave activity have also been associated with an increased perception of calmness.

3.2.5. Beta waves

Unlike alpha waves, beta waves, at the frequency of 12–30 Hz [34], occur during a heightened state of awareness. The beta state is a type of brain oscillation that occurs as a task is being completed and throughout active concentration. Beta waves are also present throughout the motor cortex during isotonic contractions and slow movements, and similar to alpha waves, are also detected by EEG. Beta activation is correlated with gains in academic performance. Increased beta activation has been associated with high arithmetic calculation ability [58]. Similarly, greater existing beta wave activity has also been associated with a higher grade point average among students. These findings may indicate an association between beta waves and increases in cognitive skills. Along with cognition, beta waves have also been shown to affect mood and emotions. For example, transcranial magnetic beta wave stimulation has shown a significant decrease in emotional exhaustion and state

Table 2
EEG changes in meditation.

Author	Style	sub	Feature-extraction	Results
Sobolewski et al. [61]	Buddhist	26	–	Event related potential.
Laxmi Shaw et al. [114]	Kriya yoga	23	Statistical features	Relative band energy is higher in δ wave for meditators.
Chang et al. [63]	Zen	41	Wavelet coefficients	Suppression in α activity
Huang et al. [64]	Zen	23	–	Rise in front θ and β
Kubota et al. [115]	So Soku	25	Spectral analysis	Increase in frontal θ
Lo and Zhu [65]	Chan	16	Wavelet-decomposition, fuzzy C-means	reflecting calm mind
Cahn et al. [72]	Mindfulness Meditation	16	Independent component analysis	(i) Reduced frontal alpha (ii) Rise in frontal theta and gamma (reflecting better synchronized function)
Ahani et al. [36]	Mindfulness meditation	34	Phase and Spectral analysis	(i) Rise in frontal θ (ii) Minor decrease in right temporal and occipital α (iii) Rise in β and EEG synchrony
Chan et al. [73]	Tibet	19	Fast fourier transform	(i) Rise in α asymmetry index (ii) increase Fm Θ
Travis [77]	Trans Meditation	26	Spectral analysis and coherence analysis	(i) increase frontal α and β (ii) unchanged coherence
Hebert et al. [81]	Trans Meditation	27	Time domain method	Rise in the α synchrony
Eskandari and Erfanian [82]	Trans Meditation	10	Wavelet decomposition	(i) α Event-Related Synchronization (ERS) and β ERS in the hand movement (ii) α event-related de-synchronization in the rest
Kjaer et al. [83]	Yoga-Nidra meditation	8	Spectral-analysis	Rise in θ power and decrease α power.

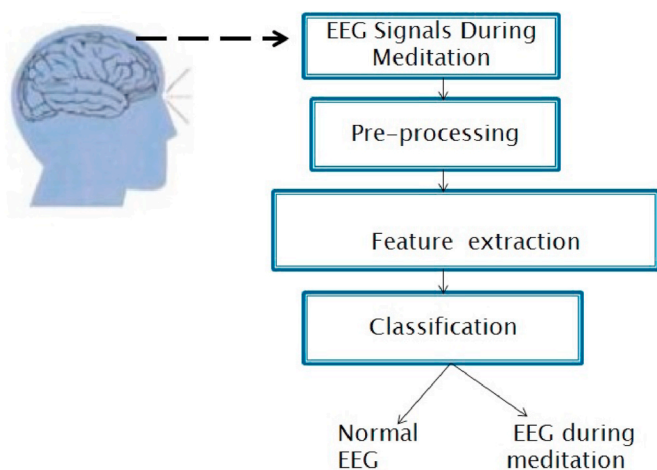


Fig. 3. EEG classification.

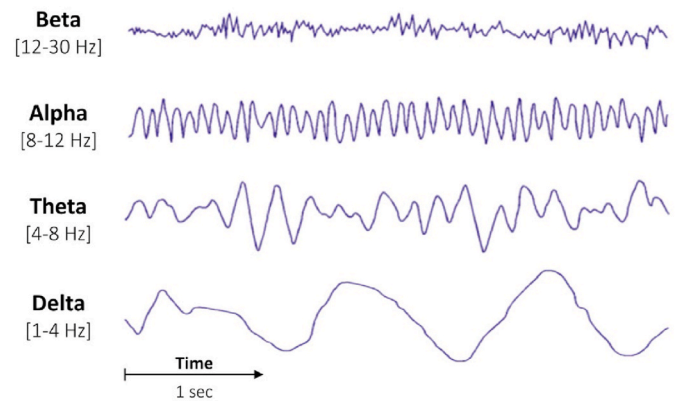


Fig. 4. Brain wave patterns.

anxiety [59]. Another study reported reduced feelings of fatigue after seven weeks of beta stimulation [34].

3.2.6. Gamma waves

Gamma waves are brain oscillations that occur at the frequency of 30–100 Hz [34]. It has been thought that these brain waves are involved in conscious attention. Previous research has observed that gamma waves originate from thalamus and will move anteriorly as they are activated to synchronize neuronal activity. Along with this, gamma waves at the frequency of 40 Hz are involved with establishing neuronal circuitry. It has been noted that the absence of this wave, often times as a result of the thalamic injury, conscious awareness can no longer form, and the individual slips into a profound coma. Gamma waves have been induced in a mice population by stimulating a set of fast-spiking inter-neurons [34].

4. Results

4.1. Meditation styles

A detailed quantitative analysis of different meditation practices are as shown in Fig. 5 below.

4.1.1. Buddhist meditation

Buddhists use meditation to remodel themselves to move beyond the disturbances of the outside world to observe the true nature of things. In Buddhist meditation, high voltage Gamma waves have been found on the EEG [60]. A randomized design study of Buddhist meditation on emotional regulation has been conducted [61].

4.1.2. Zen meditation

Zen-meditation, which is one class of Buddhist meditation, is categorized by a slow increase in alpha wave power and fast theta wave power, reflecting more internal attention and showing increased mindfulness in front region [62,63]. Long term meditators show an increase in alpha wave voltage in the frontal area and beta rise over occipital regions [64].

4.1.3. CHAN meditation

Researchers correlated occipital alpha wave rise during eyes-closed relaxation and frontal alpha wave (representing mindfulness) during CHAN meditation. A system was used for microstate analysis [65] to investigate the spatial-temporal properties of different alpha maps. It showed more focus compared to control groups. Another study showed that better cortical interactions are possessed during CHAN meditation and Chakra focusing practice than the control group [66]. In this study, interactions among various brain networks under alpha oscillations have been explored using continuous-time wavelet transformation. This

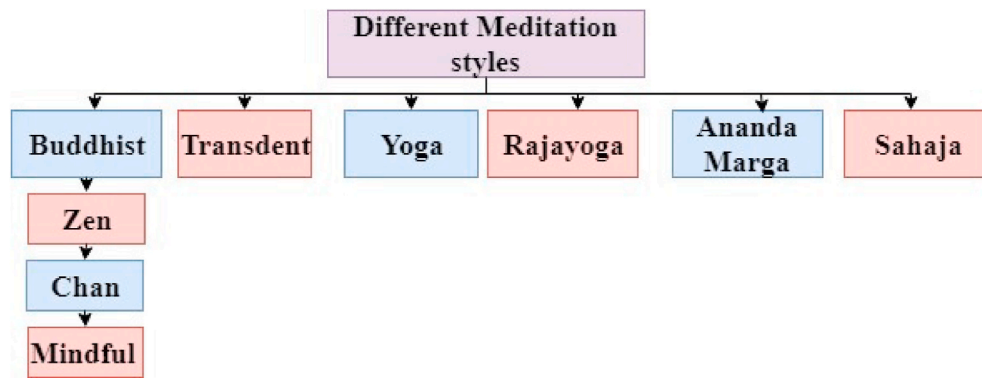


Fig. 5. Different meditation Styles.

nonlinear independent analysis identified the dominant alpha epochs in the right and left temporal regions.

4.1.4. Mindfulness meditation (MM)

Physiological excitement and distraction from unfocused ideas have been decreased in meditators. It can be used for Attention Deficit Hyperactivity Disorder (ADHD) treatment [67]. From previous phases in self-referential processing, MM can cause neuroplasticity and increase attention to inner and external stimulation. Self-referential processing is associated with the Default Mode Network (DMN) [68–72]. This research theorized how different MM states and traits impact. Cahn et al. [73] evaluated the effect of Vipassana on the frontal areas and noted reduced delta and a relative increase in theta. However, other usual modifications was not observed over alpha, beta, and frequency bands. Understandably, the primary restriction of the research was the unspecific impact on distinct frequencies of meditation exercise [72].

The meditation study on MM developed by Ahani et al. [36] by observing the spectral evaluation of registered EEG. They confirmed alpha and theta increase. Although the control group was used to analyze this randomized controlled trial, the study was limited only to novice meditators. Other IBMT (Integrative Body-Mind Training), TBRT (Triarchic Body Pathway Relaxation), and MM techniques have been studied. Still very little information is available and large scale considerations of these techniques are required for expert meditators [73,74].

4.1.5. Transcendental meditation (TM)

Different physiological and neural impacts have been reported in long-term Jacobson's Progressive-Relaxation meditators (a classic relaxation technique), TM meditators, and Progressive-Relaxation group of starters [75]. In all three, rare theta activity (5–7 Hz) has been noted. A thorough study on TM then described its effects that could be used for different clinical situations [76]. Increased frontal alpha and beta voltages and no change in coherency was found by comparing TM-Sidhi with TM program. Increased alpha and beta, respectively, represent automatic self-transcending and extensive surveillance. This analysis was performed for the frontal and parietal regions. No important differences have been noted for temporal amplitude averages [77]. It was also shown that TM could be used to reduce stress and improve ADHD symptoms [78,79]. In another research study the impact of meditation on individuals and groups was evaluated [80]. Strong delta coherency and theta coherency were observed during individual meditation with closed eyes, and no alpha coherency was noted. Distinct changes in alpha and theta coherence during group meditation show more relaxation reflected by subjects and less delta activity. TM's impacts on EEG alpha stage synchrony were explored in another research study [81]. A time-domain technique was used to observe an increase in synchrony around frontal and occipital lobes. It accounts for better-focused use of this style of meditation processes. The EEG signal analysis of event-related spectral disturbance during TM exercise was provided to

analyze brain-computer interaction systems mind controllability [82]. No control group was utilized for the investigation.

4.1.6. Yoga meditation

During Yoga Nidra and Sahaja Yoga, evidence indicates increased theta EEG activity, but more research with effects on control groups is required [83,84]. Long-Term Meditators (LTM) showed greater theta and alpha energy during Sahaja Yoga, and alpha desynchronization [85] was shown by Short-Term Meditators (STM). EEG mechanisms during Sahaja Samadhi (concentrative) meditation represent a rise in frontal region theta strength and theta consistency during profound meditation [86]. A comparison with the control group is needed again. EEG pattern of a Kundalini Yoga master's EEG pattern and Ananda Marga were theorized as reflecting more alpha and gamma group [87–90]. In view of the skill levels, Very few studies have been performed with different skill levels of practitioners of yoga meditation.

4.1.7. Other meditation practices

Paced Breathing (PB), a type of Su-Soku meditation, is a voluntary technique of breathing. It is shown to cause changes in EEG such as increase in alpha power at low and high frequency and decrease in theta power. Compared to other meditative states, PB is beneficial [91]. An EEG oscillating mechanism has been researched [92] during sophisticated breathing meditation, which is a new area to be explored. This finding endorsed the fact that the internalized attention is continually improved and is notable as indicated by an increase in theta operations, while relaxation is prominent only after a profound meditation stage as indicated by alpha activity modifications [92].

Comparisons with the rest group have been suggested as shown in Fig. 6, significant effects of both the meditation styles on the state. It accounts for their clinical use in early depressed patients, although different constraints were noted in the research, such as non-specific variables, tiny sample size, and novice subjects [28].

4.2. Feature extraction

The field of EEG analysis research is enhanced by signal processing techniques [93]. The scalp electrodes, which reflect the brain activity in terms of electrical potential, offer signals that can be processed using algorithms for signal processing to evaluate various mental operations and disclose cognitive functions. Using distinct EEG patterns from electromagnetic field activity [94,95], the inner language of the mind can be understood. EEG involves signals that are related to consciousness, motivation, and cognitive load state [96–98]. EEG information or output presents as delta, theta, alpha, beta and gamma waveforms as previously described [99].

Travis and Shear [100] made an attempt to characterize EEG signatures for distinct kinds of meditation. They did not assign ratings but appreciated the nature of meditation styles. In order to categorize the

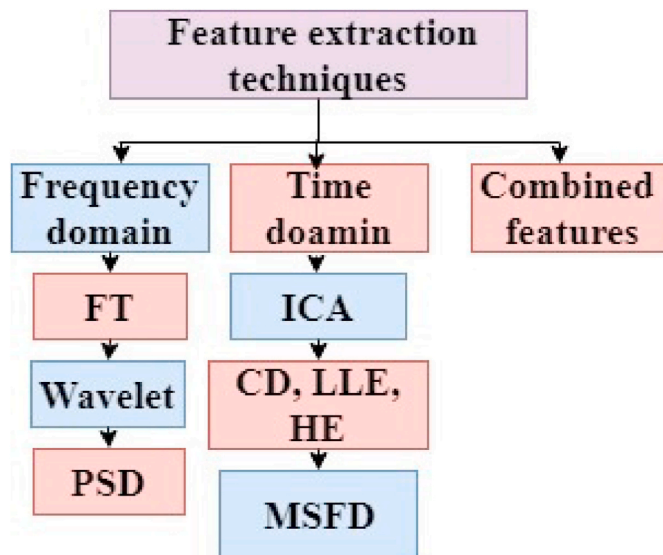


Fig. 6. Feature extraction techniques.

EEG signatures corresponding to distinct meditation methods, a well designed study is still needed [101]. It is a very tedious task to observe the nonlinear and non-stationary EEG directly [102]. Thus different linear and nonlinear methods of signal processing and their correlation with physiology have been suggested as shown in Fig. 6. Feature extraction algorithms from the pre-processed raw signal acquire spectral data and analysis of time-frequency is useful in clarifying EEG rhythmic data. It is also possible to use methods to assess consistency. Spectral covariance or coherence includes the measurement of phase regularity between signal pairs in each frequency band [103]. Higher alpha-theta coherence has been recognized as an intra and inter-hemispheric meditation capacity characteristic [104]. Since coherence cannot separate data as amplitude and phase data when two signals are related, it only measures round about the locking stage. The method of synchrony is used instead of analyzing the spectrum or coherence. It is the quantification of the degree of locking stage between distinct signals of the narrow-band [105]. Linear techniques used include Independent Component Analysis (ICA), Common Spatial Patterns (CSP), Linear Discriminator (LD) and the linear method of forecast [106]. In other words, a wide variety of algorithms extract data from EEG signals, such as Fast Fourier Transform algorithms, Hilbert-Huang Transform, Wavelet Transform, rule-based expert systems, and many other algorithms to define the temporal extents [107,108]. A technique based on Period Based Frequency Tracker (PBFT) has been suggested to maintain track of alpha frequency rhythmic differences [109]. Although Short Time Fourier Transform (STFT) can evaluate such spectral variants, this technique is better in terms of fairer frequency resolution. Also, EEG's asymmetric periodicity issue was reliably and effectively resolved by such techniques. Based on multiple parameters, for instance, Correlation Dimension (CD), Largest Lyapunov Exponent (LLE), and Hurst Exponent (HE), the intrinsic in-homogeneous features and multi nature are addressed using nonlinear signal processing methods. For the Compressed Spectral Array (CSA), the running fractal dimension, and the running attractor dimension [110], four-channel EEG analysis has been utilized. CSA's characteristics are interesting. The running attractor is useful in evaluating neural dynamics. Another region of study to explain EEG's multi-scale spatial patterns [111] is the Multi-Scale Fractal Dimension (MSFD). Nonlinear methods are more advisable than time domain [37]. A comparative analysis of different feature extraction techniques is shown in Table 2.

Signal processing can add to this field of study. EEG is the best diagnostic tool to provide secure information about brain activities and temporal dynamics related to these activities within millisecond range,

but if analyzing technique is weak, it could result in misinterpretation and may ignore specific neural correlates [102]. So it can be said that EEG is misjudged profoundly. More processing tools and engineering approaches are required to be investigated for exploring EEG information [112]. Also, meditation effects on the brain activity measured by EEG could be contaminated by the electro muscular artifacts. EEG rhythms show six times less power in 25–30 Hz band and 100 times less 40–100 Hz power in paralyzed subjects [113]. So muscle contamination is an essential issue in defining gamma EEG during meditation.

4.3. Classification

Classification can be defined as the process of segregating signals into different classes. Various popular methods used for categorization of EEG signals are K-Nearest Neighbors (KNN), Support Vector Machine (SVM) [116], Random forest, Fuzzy logic, Neural Networks [117] and Convolutional Neural Networks as shown in Fig. 7.

Sharma et al. [118] used EEG signals before and after Yoga and Sudarshan kriya meditation. Features extraction and classification were performed on these signals using Discrete wavelet transform and neural networks.

Laxmi Shaw et al. [114] collected EEG data from 23 meditators (kriya yoga) and non-meditators. Statistical features (variance, kurtosis, relative band energy, Shannon entropy, and Renyi entropy) are calculated and classified using Principal Component Analysis (PCA) clustering technique. Chuan-Yi Liu et al. [119] gathered EEG data during ZEN meditation feature extraction, and classification was performed using Wavelet transform and Fuzzy C-means classification algorithm. From results, it was observed that alpha power in the frontal region of meditator's increased. Asieh Ahani et al. [36] acquired EEG data of 34 meditators, and the collected data were analyzed using spectral and phase analysis. The SVM classifier showed good performance accuracy on this data-set. Lee et al. [37] proposed two artificial intelligence techniques (Neural Networks and SVM) for the classification of EEG during meditation by taking three statistical features. Thilo Hinterberger et al. [38] utilized 64 channel EEG of 49 people and estimated power spectral density and then classified using multiclass linear discriminant analysis classifier. Narendra Jadhav et al. [25] acquired EEG during four emotions Happy, Angry, Sad, and Relax pre and post-meditation state

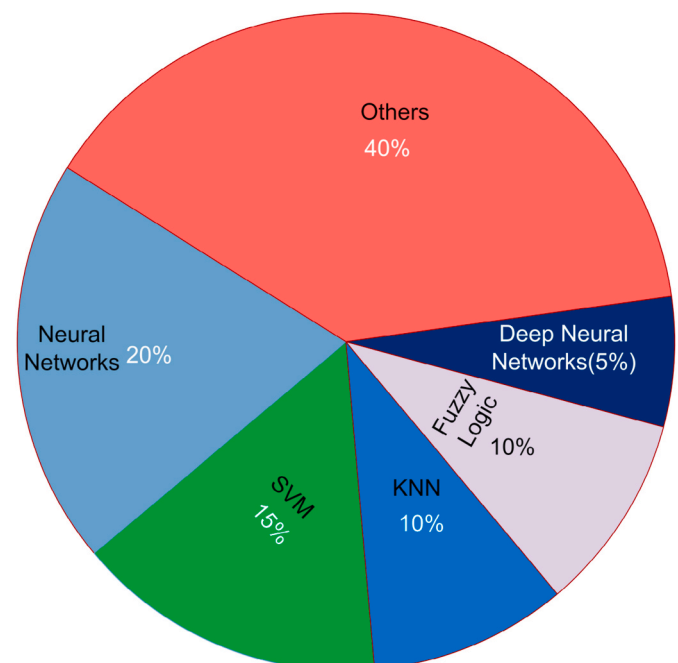


Fig. 7. Research using different classifiers.

and then classified these signals using KNN classifier. Dissanayaka et al. [120] observed EEG signals during wakeful, meditative and attentive states, and drowsiness. Spectral coherence was calculated using minimum variance distortion-less response and Welch methods. Chang et al. [41] collected 8 EEG channel data during Zen Buddhist meditation and extracted features using a wavelet transform. Then they classified these features using the Fuzzy C-means classification algorithm. Manish Sagar [42] et al. used second-order blind source identification to separate signals. Multi-tapered power spectral density estimation method was used to extract features, and a novel semi-automatic artifact removal tool was used for the classification. Atefeh Goshvarpour [43] et al. used wavelet transform and correlation dimension to extract the features, and classified them using Parzen classifier and Fisher discriminant.

In this paper, a broad spectrum of neural mechanisms under a variety of meditation styles has been reviewed. A detailed analysis of various mental states using Zen, CHAN, mindfulness, TM, Rajayoga, Kundalini, Yoga, and other meditation styles has been described by means of EEG bands. Classification of mental states using KNN, SVM, Random forest, Fuzzy logic, neural networks, Convolutional Neural Networks has been described. Classification algorithms help to predict the qualitative features of a subject's mental state by extracting useful information from the EEG data. Superior research is still required to classify the EEG signatures corresponding to different mental states. It has been concluded that increased independent brain processes are observed compared to task-free resting during meditation. More rigorous studies are required with better design, considering client variables like personality characteristics to avoid risks, randomized controlled trials, long-term effects, and large sample sizes. More rigorous clinical trials that concentrate on the use of meditation are also required. Investigations on further activities can add neuroanatomy to explore the mechanisms of mind and body. Signal processing aspects in the field of cognitive neuroplasticity have also been discussed.

5. Conclusion

Studies based on EEG brainwave signals can help medical practitioners to check the activity level of the brain, and based on the health state, different meditation practices can be applied to progress mental fitness. Yoga has advantageous effects on brain activity as reflected by changes in alpha, beta, and theta brainwaves related to enhancements in insight, memory, mind-set, and reduction in uneasiness or anxiety. Yoga is related to diminished amygdala activation and diminished negative feelings in response to emotionally disturbing pictures. Yoga also constructively affects the life systems of the mind. The use of yoga in the clinical treatment of certain neurological and psychosocial conditions is helpful to persons afflicted by these due to its potential neuroplastic effects. Future research should concentrate on yoga interventions specific to neurological and psychiatric conditions. As a result of the wide assortment of yogic practices, research should look at changing styles of yoga and evaluate the degree of their neurocognitive and neuroanatomical effects for the neurologically affected population. Investigating the specific components through which yoga enhances certain brainwave patterns, induced incremental changes and activation in certain regions of the cerebral cortex, potentially produce valuable anatomical changes in the brain is needed. With such new knowledge about specific yoga programs or modules that targets explicit neurological dysfunction and illnesses and work towards improving the manifestations (symptoms, impairments, disabilities), even if it does not address the ultimate underlying cause of the disease.

This study concentrated on the latest trends in human brain activity pattern extraction and identification during yoga and meditation. It was found that different methodologies and approaches have been used and these are presented in this article, with the aim of actually adapting such methods to clinical use. This review also brings attention to the successful achievement of high classification accuracy between yoga/

meditation induced states and normal state.

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